

Part A: Reddit Post Collector

Github: <https://github.com/lindsayadams2552/CS172-Project.git>

I. Collaboration Details: Description of Contribution of Each Team Member.

Lindsay Adams: Brainstorming, Comment Crawler, Managing File Size, Permalinks, Implemented Half of Multithreading, Error Handling, Testing, Report

Ananya Sood: Brainstorming, Embedded Links, Managing File Size, Testing, Managing Duplicates, Implemented Half of Multithreading, Report

Selina Wu: Brainstorming, Reddit API, PRAW Initialization, Dictionary, BeautifulSoup, Configured for Multiple Subreddits, Output JSON Folder, Report

Nada Salib: Brainstorming, Reddit API, PRAW Initialization, Code Framework, Search Engine Mockup Design, Testing, Report

Rhea Niraj: Brainstorming, Search Engine Mockup Design, Report

Subreddits Crawled for 500+ MB: ["movies", "television", "NetflixBestOf", "popheads", "hiphopheads", "Music", "gaming", "PS5", "NintendoSwitch", "popculturechat", "OutOfTheLoop", "comicbooks", "anime", "MarvelStudios", "StarWars", "TheLastOfUs", "marvelstudiosspoilers", "televisioncriticism", "moviescirclejerk", "MusicDiscussion", "KDRAMA", "funny", "reactiongifs", "90s", "80s", "Documentaries", "entertainment", "socialmedia", "rap", "literature", "popculture", "celebrities", "Oscars", "radio", "netflix", "indieheads", "podcasts", "InfluencerSnark", "LetsTalkMusic", "truegaming", "Fantasy", "oscarace", "grammys", "Metal", "EDM", "Jazz", "rock", "books", "pcgaming", "Games", "PS4", "halo", "Hulu", "HBOMAX", "animesuggest", "horrorlit", "InternetIsBeautiful", "startrek", "booksuggestions", "MovieSuggestions", "bestofnetflix", "thesopranos", "SubredditDrama", "Suggestmeabook", "riverdale", "TheTraitors", "olympics", "RedditWritesSeinfeld", "spotify", "TheTraitors", "TrueCrimePodcasts", "tolkienfans", "MuseumOfReddit", "tldr", "kpophelp", "asoiaf", "scrubs", "WikiLeaks", "serialpodcast", "AskReddit", "wikipedia", "discordapp", "Shoestring", "HobbyDrama", "politics"]

II. Overview of system, including (but not limited to)

1. Motivation/End Goal of the Search Engine

The end goal of our search engine is to get information regarding the theme of pop culture in multiple subreddits. We chose to pick pop culture because it is a very broad field that covers many different entities such as movies, television, gaming, music, celebs and other relevant subreddits. Additionally, since our search engine applies to our theme with 500MB of raw data, our vast subreddit will allow for a diverse search engine. Because we are pulling from subreddits in real-time, we can collect valuable information that can be easily searchable through our search engine. We focused on the most popular subreddits within pop culture themes so that we could create a specialized search engine that allows users to find relevant reddit posts, opinions, and discussions.

Our motivation for creating this search engine is to ensure that all of the unstructured information in the theme of pop culture can be easily accessible. We feel like pop culture discussions are often buried across so many different subreddits, posts, and comment sections.

The search engine tool gets rid of the time consuming process of users having to parse through multiple communities to gather information and catch up on the newest trending topics. This search engine will allow us to allow us to index all of this information and organize and filter it based on categories so that users can access more targeted searches and access high quality content with minimal effort.

- [User Interface Concept Design](#) [links to Canva]

2. Architecture.

a. *Authentication/Initialization of PRAW*

- The first step in our system architecture is initializing PRAW to make sure that our program is authenticated with Reddit's servers. We are using a registered Reddit app's client ID, client secret, and user agent so that our crawler is identified properly. The credentials are passed through the PRAW constructor so that the authorized requests can be sent to Reddit's API so that Reddit is able to track our API usage. Additionally, the authentication allows our crawler to interact with Reddit and see the subreddit streams and pull posts, comments, and data in the external links. Overall, the initialization of PRAW allows us to crawl and collect data.

b. *Subreddit Crawler*

- The next major part of our architecture is the subreddit crawler. We have a list of seed subreddits that are related to pop culture and for each subreddit we access four integral streams ("top," "hot," "new," and "rising"). Through this we are able to gather the most relevant and upvoted posts. The crawler we implemented iterates through these streams and pulls valuable data ("title," "body," "author," "upvote," "postID," "postImage," "postURL," "comments," "category").
- To make sure that our crawler is optimized and efficient, we kept track of the postIDs that we have already crawled so that duplication does not occur when we are crawling multiple streams from the same subreddit. We have a total data collection goal of 500MB broken into 10MB per file.

c. *BeautifulSoup to Extract HTML Titles*

- Another crucial part of our crawling process is making sure that we handle posts that contain external links. We use the BeautifulSoup library to fetch and parse the linked HTML pages. If a reddit post has a URL, we make an HTTP request to that page through the requests library and if that content is an HTML page, we use BeautifulSoup to extract the <title> tag and attach it to the Reddit post data. We connect to more broad information sources, and extracting the title allows for less slowdown for our crawler.

d. *Multithreading/Parallelism*

- To make sure that the performance of our crawler is optimized, we implemented multithreading through python's ThreadPoolExecutor. Our system submits the I/O tasks to a thread pool that will handle the operations in parallel. Crawling speed is improved greatly because the thread pool creates 3 worker threads, improving our throughput without overwhelming Reddit. Since each post's comment fetching and external link extraction are separate background tasks that are

collected later when we write the data into the JSON files we use the available CPU and network resources, multithreading allows us to gather 500MB of data.

e. *JSON Files and Organization of Data*

- i. The last step of our architecture is storing and organizing the data that we collected. We put each post's information into a JSON format that is written line-by-line into the .json files. Each output file does not exceed 10MB, and after it completes a file, it starts writing to a new one. The file organization structure allows the dataset to be easily managed, loaded, and indexed for the next phase of our project.

3. Duplication Handling Strategy

Our crawler handles duplicate posts by having a `seen_posts` set that is maintained as we are crawling. When a new reddit post is crawled, its unique post's ID is checked against the `seen_posts`, and if seen, the crawler will skip over that post. This helps us make sure that even when we are crawling over different streams that could have the same post, we do not process the post more than once.

This duplicate handling strategy that we implemented allows us to improve the accuracy of the data that we collect and allows us to not store any data that is redundant. If we did not have this duplication handling strategy in our code, we would help up by having JSON files that have repeated entries, thus making our data limit overinflated. It is ideal to make sure that each post is only being collected once so that that crawler contains accurate information that can accurately represent the subreddits that we are crawling.

4. The Crawling or Data Collection Strategy.

Our data collection strategy utilized Reddit's PRAW API to crawl reddit posts and comments, as well as embedded HTML links. We accessed Reddit's API with a script application, so we created the values *client ID* and *client secret*. These two values, along with our *user agent* for sourcing network requests, are initialized and authenticated in the crawler with the *reddit* object.

Because we need to collect 500MB of data total, we crawled multiple subreddits as seeds related to the topic of pop culture. We used a general topic like pop culture to include as many subreddits as possible; they span from areas like celebrity gossip to fandom and media. These subreddits are stored in a list called *subreddits*. We loop through the defined subreddits in the list to crawl for data; each post has comments and we crawl through all of them as well. To calculate the file and data size, we implement a file index and establish the maximum size per file (~10MB) and the total data limit (500MB). We also keep track of the current size and total size of the files. The data is stored in JSON files in the directory *redditFiles*, with a post per row.

Since there might be duplicate posts, a set of seen posts is created to filter out duplicates. The posts are then crawled by top, hot, new, and rising, which are labels paired with PRAW. The post information collected throughout is organized in a dictionary called *data* with values including the subreddit name, post title and text, post author, upvotes, post ID and URL, label and category, and comments, which are stored in a list. If a Reddit post contains a URL to an HTML page, we use BeautifulSoup to fetch and save the page title, cleaning it if necessary. Non-HTML content is skipped otherwise. In situations where there's a self-post, we crawl the Reddit

permalink as well. The comments and link data are saved and written to the files accordingly. The collective data is encoded in the files as UTF-8 once they're crawled. The JSON files are managed so that when one exceeds 10MB, we close the current file and open a new one until we reach 500MB. The file index is incremented so the crawler knows how many files have been created.

Since there are so many different subreddits to go through, we also implemented multi-threading to speed up the process and maximize the CPU power. The created thread pool is useful for parallel HTML and comment fetching since we're parsing through every link and comment. This makes it easier to continuously add subreddits to crawl to reach the 500MB requirement.

5. Data Structures Employed.

- a. **Dictionary**: We used a dictionary to collect and organize information pertaining to each individual Reddit post. It first stores the name of the subreddit that the post belongs to. Next, it captures the title, body text (HTML cleaned), and the author's username. Each Reddit post's upvote count, which reflects its popularity and visibility within the subreddit, is also stored. In addition, the dictionary records the post's unique Reddit ID and its direct permalink. The dictionary has a "comments" placeholder, which is initially empty and fills as the comments get crawled later in our code. Lastly, there is a category section which stores which stream label the Reddit post belongs to, such as top, hot, new, and rising. By using a dictionary, our code is made more modular and organized, allowing us to store, update, and retrieve information for each reddit post easily. The dictionary is also used to populate our JSON file, where each post's information is stored with its relevant fields.
- b. **Sets**: We used a set to prevent crawling duplicate links. We implemented the set within a loop, populating it with each post's unique ID. Every time a new Reddit post is encountered, we check whether its ID already exists in the set. If it does exist, the post is skipped, otherwise it stores the new ID. This approach allows us to effectively avoid duplicates crawling across stream labels (top, hot, new, and rising).
- c. **Lists**: Lists were used for multiple different purposes within our code for easier storage, updating, and retrieving of information. We first used lists to store the names of the subreddits to be crawled. We then used them to store our stream labels (top, hot, new, and rising), and pair them with PRAW's subreddit streams during crawling. The comments from each post are also stored within a list since there can be multiple comments per post. Lastly, we use a list (future_to_data) to temporarily store all posts whose comments and link titles are still being fetched through multithreading.
- d. **Strings**: Strings are used all throughout our code to represent various textual data. Items such as subreddit names (e.g. "popculture", "movies", "television") and stream labels ("top", "hot", "new", "rising") are stored into lists and utilized for crawling. Our error, completion, and current MB messages are also strings. Our JSON file also is transformed into a string before being written to the file. Many other instances of strings occur in our code, all of which improve communication, retrieval, and storing of information.

III. Limitations (if any) of the System.

The primary limitation is the speed of our system, although we did our best to reduce this through multi-threading. However, the API rate limit of PRAW can slow down the crawler because we're going through so many subreddits. Retrieving the HTML titles using BeautifulSoup also produces timeout errors after 10 seconds and we don't crawl images. Another limitation in retrieving the HTML links is that we only take the titles of the embedded links, not the actual page content. Finally, because of the permissions of our client credentials, we can't access any private subreddits and can't take usable data from deleted post authors.

IV. Instruction on how to deploy the system. Ideally, you should include a crawler.bat (Windows) or crawler.sh (Unix/Linux) executable file that takes as input all necessary parameters.

install python3

install python libraries: pip install praw requests beautifulsoup4

change directory into CS172-Project: cd CS172-Project

run crawler: python3 PRAW.py

V. Screenshots Showing the System in Action.

```
Downloaded 37.4 MB...
Downloaded 37.5 MB...
Downloaded 37.7 MB...
Downloaded 37.8 MB...
Skipped non-HTML content at https://imgur.com/XJT8g9Q.jpeg
Downloaded 37.9 MB...
Downloaded 38.0 MB...
Skipped non-HTML content at https://i.imgur.com/zPYQjcu.jpg
Downloaded 38.1 MB...
Downloaded 38.3 MB...
Downloaded 38.4 MB...
Downloaded 38.5 MB...
Error fetching https://www.cbr.com/tiger-king-baskins-husbands-will-forged/: ('Connection aborted.', RemoteDisconnected('Remote end closed connection without response'))
Skipped non-HTML content at https://imgur.com/L1mnGEL.jpg
Downloaded 38.7 MB...
Downloaded 38.8 MB...
Downloaded 38.9 MB...
Downloaded 39.0 MB...
Downloaded 39.2 MB...
Downloaded 39.3 MB...
Downloaded 39.4 MB...
Downloaded 39.5 MB...
```

```
426 {"subreddit": "celebrities", "title": "Selena Gomez", "body": "", "author": "Denzell1999", "upvotes": 858,
    "postID": "1kbu445", "postImage": "https://www.reddit.com/gallery/1kbu445", "postUrl": "/r/celebrities/comments/1kbu445/selena_gomez/", "comments": [{"Benny Blanco is the most fortunate Hobbit in all of middle earth.",
    "Amazing set you brought to us, Selena is magnificent!", "Benny Blanco is the man I strive to be", "Total smoke show.", "Damn. That thickness.", "Mother of god.", "Remember when she didn't even gave credit to the person who
    donated her kidney?? From that moment onwards I started hating her.", "I preferred the \u201cbefore\u201d
    version\u2026before the freakishly giant jugs.", "She must have had a boob job!", "Always was a
    crush from Wizards of waverly place", "Holy hell she keeps getting better", "Benny Blanco must have the
    greatest personality ever", "Heavies", "Here hottest video was when she cried about the illegals being
    deported", "Is it wrong that I love seeing the veins in those massive flesh bags?", "Big ol mommy tits", "Noods
    already!! How long do we have to wait to see that fat puss?", "This", "This.", "What did you just try to say?",
    "That was my favorite too!", "Donated*", "I pulled a back muscle from laughing so hard at that video"],
    "category": "hot", "linkTitle": "Reddit - The heart of the internet"}
427 {"subreddit": "celebrities", "title": "Diddy 'Is Very Eager' to Testify at His Sex Abuse Trial (Exclusive)",
    "body": "", "author": "ElvisIsNotDjed", "upvotes": 0, "postID": "1kcwsst", "postImage": "https://people.com/
    diddy-very-eager-testify-sex-abuse-trial-exclusive-11724716", "postUrl": "/r/celebrities/comments/1kcwsst/
    diddy_is_very_eager_to_testify_at_his_sex_abuse/", "comments": [], "category": "hot", "linkTitle": "Diddy 'Is
    Very Eager' to Testify at His Sex Abuse Trial (Exclusive)"}
428 {"subreddit": "celebrities", "title": "\u2018I Kissed A Girl\u2019 Singer Jill Sobule\u2019s Body Found After
    Mysterious House Fire", "body": "", "author": "ElvisIsNotDjed", "upvotes": 1, "postID": "1kcw1fm", "postImage":
    "https://www.boredpanda.com/singer-jill-sobule-passes-away-house-fire-accident/?utm_campaign=eind&
    utm_source=reddit&utm_medium=ref", "postUrl": "/r/celebrities/comments/1kcw1fm/
    i_kissed_a_girl_singer_jill_sobules_body_found/", "comments": [{"Astronauts did it?", "Our loss!}], "category":
    "hot", "linkTitle": "\u2018I Kissed A Girl\u2019 Singer Jill Sobule\u2019s Body Found After Mysterious House
    Fire"}
429 {"subreddit": "celebrities", "title": "Dua lipa sexy dress", "body": "", "author": "bigmax46", "upvotes": 59,
    "postID": "1kc4da5", "postImage": "https://i.redd.it/r038cpbk05ye1.jpeg", "postUrl": "/r/celebrities/comments/
    1kc4da5/dua_lipa_sexy_dress/", "comments": [{"Shortly afterward, she shed her skin and looks brighter and
    shinier."}], "category": "hot"}
```

